



Stop Wringing Your Hands Over PSSO.

Deploy Passkeys Now!

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Auth is a mess:

Passwords are hackable.

MFA is phishable.

Your Org. is vulnerable.

The solution is passkeys

- **secure by design (non-phishable)**
- **easy to use**
- **faster to use**
- **available now**

TATYN (Time to Access the Thing You Need)



Password (memorized or
copy/paste)



Password + MFA



Passkey (w/autofill)

Passkeys

- **secure FIDO2 credentials**
- **hardware-based authentication factor**
- **used to access anything protected by your identity provider (IdP)**

The future is already here.

We have all the tools we need.

Today.

(assuming your Identity Team is ready)

“Passwordless” is Microsoft's vision for enterprise authentication

Platform SSO is Apple's vision for enterprise Mac authentication

Passkeys are the junction of Apple’s and Microsoft’s roadmaps.

The future is already here.

Still using Passwords and/or MFA?

Passwords & MFA are the new *Binding*

(Somebody please quote me on that)

The future is already here.

We have all the tools we need.

We don't need Simplified Setup

Passkeys NOW!

**Let's build an SSO to passkeys
upgrade workflow**

Requirements and Challenges

- **The SSO Extension MDM Profile and the PSSO MDM Profile cannot coexist**
 - We must remove the SSO MDM profile, then deploy the Platform SSO MDM profile
- **The Mac must restart *after* the SSO Extension MDM Profile is removed and *before* the PSSO MDM Profile is deployed**
 - We cannot skip this. It removes old keychain items and cached files
- **Because everyone hates restarting, the restart *must* be *enforced***
- **Therefore, IT must automate this workflow to create the best possible UX**
- **The automation itself must span across/continue after the restart**
- **The Mac needs to be running macOS 26 Tahoe to immediately trigger the Entra ID (re-)registration**

Requirements and Challenges

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 - We cannot skip this. It removes old keychain items and cached files
 - Also Sean Rabbitt says so. Don't nobody argue with Señor Rabbitt
- **Because everyone hates restarting, the restart *must be enforced***
- **Therefore, IT must automate this workflow to create the best possible UX**
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Requirements and Challenges Opportunities

- **The SSO Extension MDM Profile and the PSSO MDM Profile cannot coexist**
 - Brilliant Idea: What if removing the SSO Extension MDM Profile could *trigger* the workflow?
- **Because everyone hates restarting, the restart *must be enforced***
 - Idea: Let's use a Swift Dialog message to give our colleagues the ability to close their work
 - Swift Dialog will remain on the screen until the user restarts
- **If the Mac is running macOS 26 Tahoe, when the PSSO MDM Profile is deployed, it will immediately trigger the Entra ID (re-)registration**
 - Brilliant. Let's leverage that!

Planning the Workflow - Components Needed Before the Restart

- Removing the SSO Extension MDM Profile from target Macs will trigger the workflow
 - We need a Smart Group to detect Macs without this profile. We'll call it "READY for PSSO Config"
- We need a Policy to ask the user to restart. We'll call it "Restart for Passkey Configuration" and set it to trigger at the next checkin. The Policy consists of:
 - a script, which we will call "restart_for_passkey_set-up.sh" which
 - ▶ displays a Swift Dialog message describing the passkey deployment steps and a button to restart.

Planning the Workflow - Components Needed After the Restart

Engineering Challenge #1 of 2:

How do we automate the deployment of the “Platform SSO Extension” MDM profile as *quickly* as possible after a restart? Let's work through this logically, but backwards.

The “Platform SSO Extension” MDM profile will be deployed to Macs in a Smart Group ([more on this later](#))

We'll need to run a recon to get Macs to join this Smart Group

We'll need a way to ensure the recon *only* runs for Macs in our workflow and because running too many recons is BAD

That recon must be triggered by a Policy for members of a (by definition) different Smart Group with the following conditions:

- a. the SSO profile has been removed
- b. the PSSO profile has not yet been deployed

Planning the Workflow - Components Needed After the Restart

Engineering Challenge #2 of 2:

If the “Platform SSO Extension” MDM profile is scoped *incorrectly*, or deploys at the *wrong* time, TWO BAD THINGS could happen:

- a. we could have an SSO profile and a PSSO profile on the Mac at the same time
- b. the Entra ID registration will kick off before we are ready

We'll need to create a Smart Group which has the following criteria:

- a. the SSO profile has been removed
- b. proof that the Macs are part of this workflow. But how?

Planning the Workflow - Components Needed **Before the Restart**

- Removing the SSO Extension MDM Profile from target Macs will trigger the workflow
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- We need a Policy to ask the user to restart. We'll call it "Restart for Passkey Configuration" and set it to trigger at the next checkin. The Policy consists of:
 - a script, which we will call "restart_for_passkey_set-up.sh" which
 - displays a Swift Dialog message describing the passkey deployment steps and a button to restart.
 - creates ~/Library/PSSO_Staging/ folder (or any "meta-tag" of your choosing) to help trigger the PSSO MDM profile deployment

Planning the Workflow - Components Needed After the Restart

Engineering Challenge #2 of 2:

If the “Platform SSO Extension” MDM profile is scoped *incorrectly*, or deploys at the *wrong* time, TWO BAD THINGS could happen:

- a. we could have an SSO profile and a PSSO profile on the Mac at the same time
- b. the Entra ID registration will kick off before we are ready

We'll need to create a Smart Group which has the following criteria:

- a. the SSO profile has been removed (we already know this)
- b. proof that the Macs are part of this workflow. But how?

We'll also need an Extension Attribute to gather this proof/meta-tag, so that the recon in Engineering Challenge #1 of 2 slide reads this proof

The Workflow - Before the Restart

1. Remove the SSO Extension MDM Profile from target Macs by adding specific groups to Scope > Exclusions (OR for Self Service version, add the Mac to a Static Group which is excluded)
2. Macs without an SSO Extension MDM Profile join the Smart Group which we'll call "READY for PSSO Config" for the sake of this demo
3. Members of "READY for PSSO Config" are eligible for the Policy we'll call "Restart for Passkey Configuration" which triggers at the next checkin. The Policy runs:
 - a. the "restart_for_passkey_set-up.sh" script which
 - i. creates the ~/Library/PSSO_Staging/ folder (or any "meta-tag" of your choosing)
 - ii. displays a Swift Dialog message describing the passkey deployment steps and the need to restart.

[the user restarts the Mac by clicking "Restart Now", the only button in the SD message]

The Workflow - After the Restart

4. After the restart, a special recon will run, as part of a Policy we'll call "Inventory Collection-STARTUP-for PSSO". During this recon,
 - a. an Extension Attribute looks for the presence of the ~/Library/PSSO_Staging/ folder
 - b. Macs which have this folder join a Smart Group we'll call "READY for PSSO MDM Profile" Smart Group to which the "AUTH-MS Platform SSO Extension" MDM profile will be deployed
5. The "AUTH-MS Platform SSO Extension" MDM profile deploys soon after reaching the Desktop
6. The deployment of the "AUTH-MS Platform SSO Extension" MDM profile triggers an Entra ID registration, which, as part of the PSSO settings, includes a passkey.
7. The user follows the prompts which are built into macOS

The Workflow - What the user sees

A Swift Dialog message notifies the user of the passkey deployment, including the need for a restart.

[the user restarts the Mac by clicking "Restart Now", the only button in the SD message]

The user is prompted to (re-)register with Entra ID (macOS notification)

Clicking “Register” prompts the user to (re-)register with Entra ID, setup a passkey, and enable auto-fill

What the user sees

Restart Required for Passkey Setup



Passkeys for macOS

Before we can generate a passkey, your Mac requires a restart.

Please save your work then click 'Restart Now'.

Thanks for keeping Sophos safe!

[Restart Now](#)



Registration Required

Please register with your identity provider.



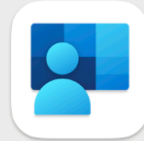
Registration Required

Please register with your identity provider

Register

IMPORTANT:

The "Register" button *will not appear* unless the user moves the cursor over the macOS notification. Make sure users and the Help Desk know this in advance.



Platform Single Sign-on Registration



Device Registration

Register your Mac computer with Sophos.

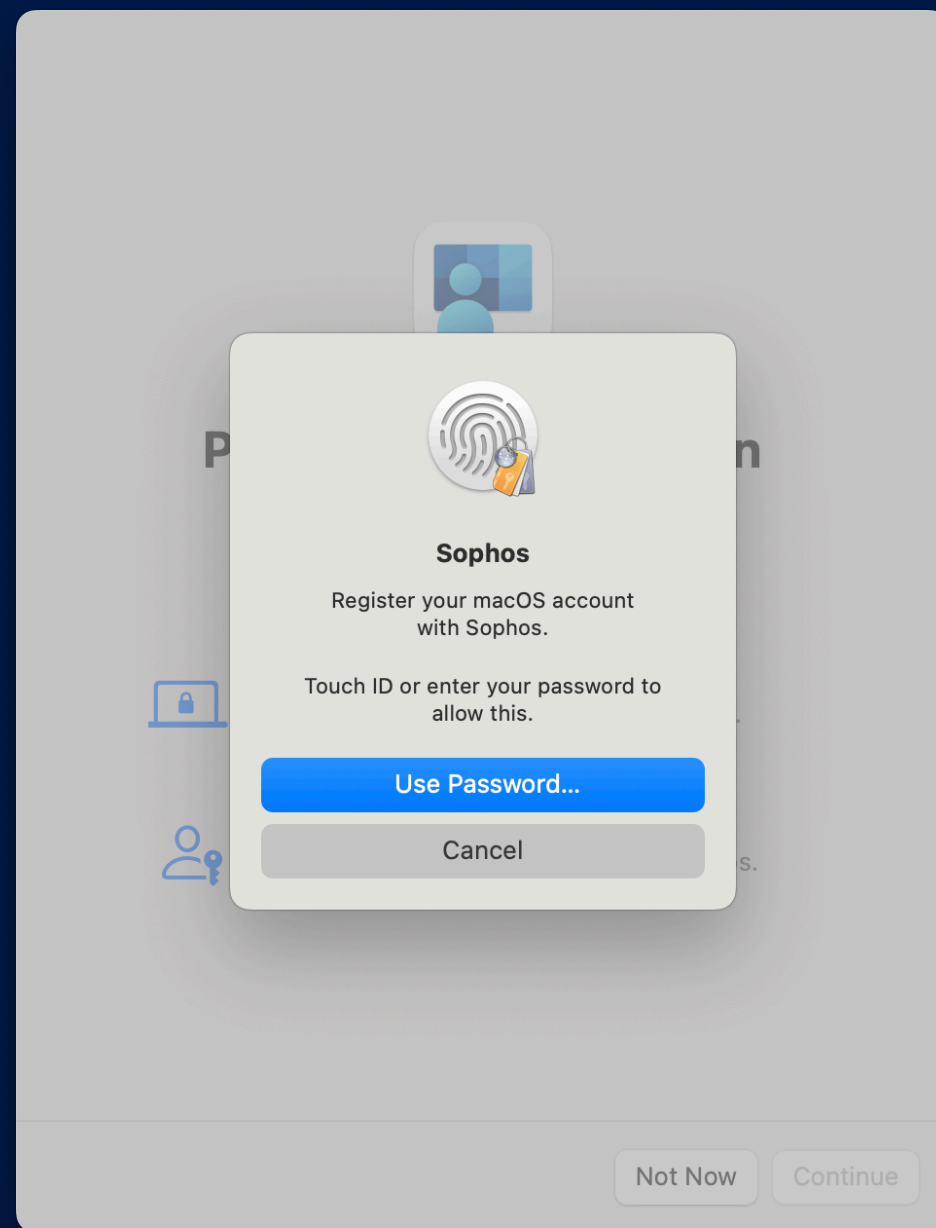


Account Registration

Register your macOS account with Sophos.

Not Now

Continue



Register your device with Microsoft Entra



Sign in

Email, phone, or Skype

No account? [Create one!](#)

[Can't access your account?](#)

Back

Next



Sign-in options

[Terms of use](#) [Privacy & cookies](#) ...

[More information](#)

Close

Registering Device...

Not Now

Continue

Register your device with Microsoft Entra

Sign In

You don't have any passwords or passkeys saved for this app. You may be able to use one of the options below to sign in.

☒  Scan QR Code

☐  Use Security key

Cancel

Continue

[Terms of use](#) [Privacy & cookies](#) ...

 [More information](#)

[Close](#)

 Registering Device...

[Not Now](#)

[Continue](#)



**Use your other device to choose a passkey for
"Company Portal".**

Cancel



Enable your Entra ID passkey

To use your Entra ID passkey, you must enable Company Portal as a Passkey Provider.

To complete this action, open the System Settings app and navigate to:
General > Autofill & Passwords > Autofill from > Enable Company Portal.

AutoFill Passwords and Passkeys

Automatically suggest passwords, passkeys, and verification codes when signing in to apps and websites.



AutoFill from



Passwords

Passkeys, passwords, and codes



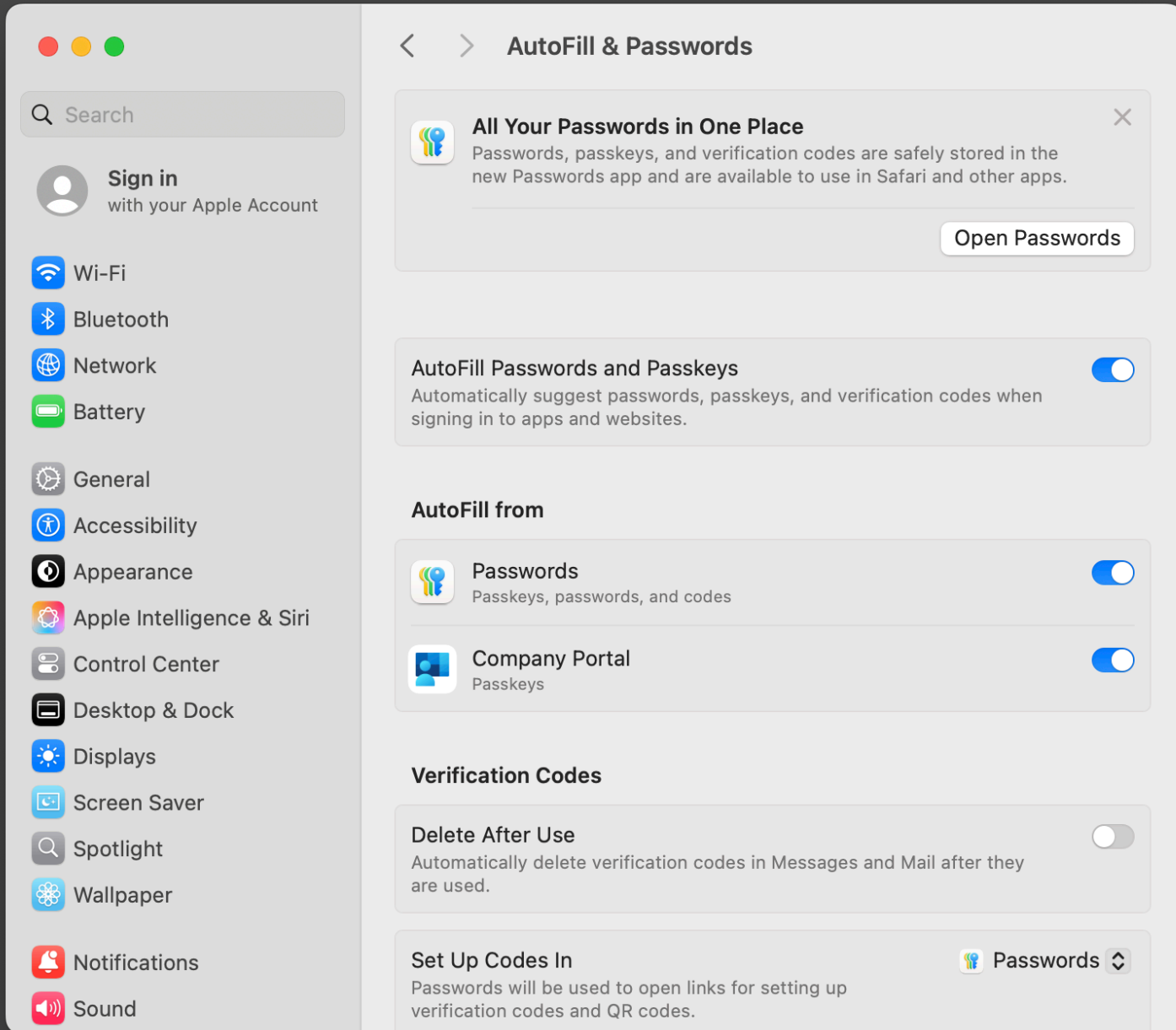
Company Portal

Passkeys



Open System Settings

Dismiss

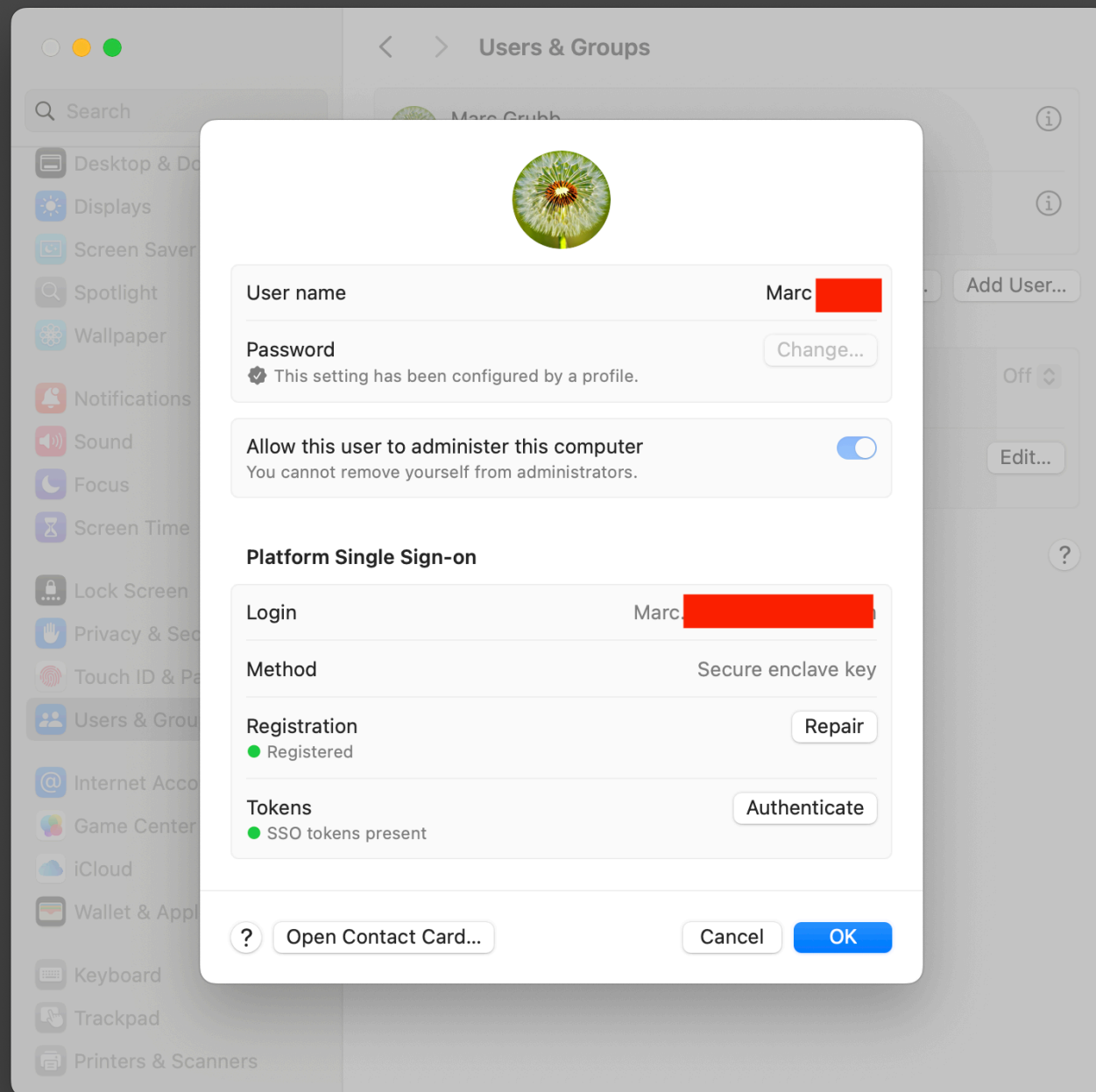




Platform Single Sign-on Registration

Registration Complete

Close



The future is already here.

What are you waiting for?

Passkeys NOW!

resources:

<https://github.com/ka-pow-er/>

[#platform-sso](#)



BONUS CONTENT

**What is the deal with
“Passwordless”?**

Macs need Passwords

**“Passwordless” is Microsoft's vision
for enterprise authentication**

**learn.microsoft.com shows 3,405
results for “Passwordless”**

Apple Platform Security Guide?

ZERO

Conclusion: "Passwordless" is a Microsoft term and has nothing to do with Macs or macOS

Does this get us anywhere?

What does Microsoft mean, when it says “Passwordless”?

**“Passwordless”
is short-hand for
(a shorter way to refer to)
“phishing-resistant
passwordless authentication”.**

**“Passwordless” is an overall
Security and Identity strategy
which specifies that *only*
phishing-resistant
passwordless authentication
is used to access
Entra ID-protected resources.**

**Passwordless does not mean
that organizations merely offer a safer,
alternative means of authentication
in addition to passwords.**

**Passwordless means
that we don't use passwords *at all*
to access protected resources
in our environments!**

(We only use phishing-resistant authentication)

Passwords don't even exist!

(Because we only use phishing-resistant authentication)

Does passwordless apply to macOS?

NO.

passwordless is a *Microsoft* term and has nothing to do with Macs or macOS

Does passwordless apply to macOS?

NO.

macOS is UNIX, and continues to require a password across the OS

So how and where do Macs, which need passwords, exist within a passwordless Microsoft Entra ID environment?

**Macs must securely access
corporate resources
protected by
An identity provider (Entra ID)

*using a phishing-resistant
passwordless authentication factor***

**we literally turn our Macs into a
phishing-resistant passwordless
authentication factor**

And we do this with...

passkeys!

**And what about syncing passwords to
Entra ID?**

**And what about syncing passwords to
Entra ID?**

Stop it!

**(You cannot sync a password which
does not exist)**

Microsoft's recommendation?

**Go back to using the MDM spec to
manage password strength**

**While you transition to fully passwordless,
stop syncing passwords over the network**

